Interpreting Real-World Graphs

Contextual Graphing Worksheet

Name: Date:

Instructions: Complete Part 1 and Part 2 below. Be sure to plot your points carefully and construct a clear graph.

Part 1: Taxi Fare (Linear Graph)

A local taxi company charges a $5 starting base fee just to get in the car, plus $2 for every mile traveled.

1. Fill in the table:

Miles (x)	Total Cost (y)
0	
1	
2	
3	
4	

2. Graph the data:

Construct a graph below. Label the x-axis "Miles" (from 0 to 5) and the y-axis "Total Cost" (from 0 to 15). Plot the points and draw a straight line through them.

3. Analysis:

- What is the y-intercept of this graph, and what does it mean in this real-world context?
- What is the rate of change (slope) of this graph?
- Represent the graph in a linear equation.

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Part 2: Ball's Height (Quadratic Graph)

A ball is thrown straight up into the air. The height of the ball (in feet) is tracked over several seconds.

1. Table of Values:

Time (s)	Height (ft)
0	0
1	48
2	64
3	48
4	0

2. Graph the data:

Construct a graph below. Label the x-axis "Time in Seconds" and the y-axis "Height in Feet". Plot the points and draw a **smooth curved parabola** connecting them. *Make sure it does not look like a straight line!*

3. Analysis:

- What is the vertex (highest point) of the graph? What does it represent in the real world?
- Where does the graph cross the x-axis (x-intercepts)? What do these points mean in this context?
- Represent the graph in a quadratic equation.

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Part 3: Savings Account (Linear Graph)

You open a new savings account with an initial deposit of $50. Each week, you deposit an additional $20.

1. Fill in the table:

Weeks (x)	Total Saved (y)
0	
1	
2	
3	
4	

2. Graph the data:

Construct a graph below. Label the x-axis "Weeks" and the y-axis "Total Saved ($)". Plot the points and draw a **straight line** through them.

3. Analysis:

- What is the y-intercept, and what does it represent in this context?
- What is the rate of change?
- Represent the graph in a linear equation.

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Part 4: Rocket Launch (Quadratic Graph)

A toy rocket is launched straight up from the ground. Its height in meters is recorded over several seconds.

1. Table of Values:

Time (s)	Height (m)
0	0
1	64
2	96
3	96
4	64
5	0

2. Graph the data:

Construct a graph below. Label the x-axis "Time (s)" and the y-axis "Height (m)". Plot the points and draw a **smooth curved parabola**.

3. Analysis:

- Between which two seconds does the rocket reach its highest point (vertex)?
- What do the x-intercepts mean in this context?
- Represent the graph in a quadratic equation.

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Part 5: Temperature Drop (Linear Graph)

A cup of hot coffee starts at 100°C. It cools down at a steady rate of 5°C every minute.

1. Fill in the table:

Minutes (x)	Temp (y)
0	
2	
4	
6	
8	

2. Graph the data:

Construct a graph below. Label the x-axis "Minutes" and the y-axis "Temperature (°C)". Plot the points and draw a **straight line** through them.

3. Analysis:

- Is the rate of change positive or negative? What does that mean?
- What is the y-intercept?
- Represent the graph in a linear equation.

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Part 6: The High Dive (Quadratic Graph)

A diver jumps off a 15-foot high diving board.

1. Table of Values:

Time (s)	Height (ft)
0	15
1	20
2	15
3	0

2. Graph the data:

Construct a graph below. Label the x-axis "Time (s)" and the y-axis "Height (ft)". Plot the points and draw a **smooth curved parabola**.

3. Analysis:

- What is the vertex of this graph? What does it represent?
- What is the starting height?
- Represent the graph in a quadratic equation.

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Part 7: Submarine Descent (Linear Graph)

A research submarine starts at sea level (0 meters) and descends at a constant rate of 20 meters per minute.

1. Fill in the table:

Time (min)	Depth (m)
0	
1	
2	
3	
4	

2. Graph the data:

Construct a graph below. Label the x-axis "Time (min)" and the y-axis "Depth (m)". Plot the points and draw a **straight line** through them.

3. Analysis:

- Why will the y-values be negative in this scenario?
- What is the rate of change?
- Represent the graph in a linear equation.

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Part 8: Basketball Free Throw (Quadratic Graph)

A basketball is shot towards the hoop.

1. Table of Values:

Distance (ft)	Height (ft)
0	6
5	12
10	14
15	12
20	6

2. Graph the data:

Construct a graph below. Label the x-axis "Distance (ft)" and the y-axis "Height (ft)". Plot the points and draw a **smooth curved parabola**.

3. Analysis:

- What is the highest point the basketball reaches?
- At what distance does it reach that highest point?
- Represent the graph in a quadratic equation.

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Part 9: Gym Membership (Linear Graph)

A local gym charges a $40 sign-up fee, plus $15 per month.

1. Fill in the table:

Months (x)	Total Cost (y)
0	
1	
2	
3	
4	

2. Graph the data:

Construct a graph below. Label the x-axis "Months" and the y-axis "Total Cost ($)".
Plot the points and draw a **straight line** through them.

3. Analysis:

- What does the y-intercept represent in this situation?
- How much would it cost after 5 months?
- Represent the graph in a linear equation.

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Part 10: Bouncing Ball (Quadratic Graph)

A golf ball is dropped from a roof and its first bounce is tracked.

1. Table of Values:

Time (s)	Height (ft)
0	0
0.5	12
1	16
1.5	12
2	0

2. Graph the data:

Construct a graph below. Label the x-axis "Time (s)" and the y-axis "Height (ft)". Plot the points and draw a **smooth curved parabola**.

3. Analysis:

- What does the x-intercept at 2 seconds represent?
- What is the maximum height of the bounce?
- Represent the graph in a quadratic equation.

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